

Low Power FSK Transceiver using ADPLL with direct modulation and integrated SPDT for BLE Application

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Abstract—This paper presents a low power FSK transceiver with ADPLL based on direct modulation and integrated SPDT switch for Bluetooth low energy application. To ensure that the proposed low power transceiver can operate at 1 Mbps data rate, FSK modulation is implemented using an all-digital phase locked-loop with direct modulation technique. The SPDT switch is integrated to share the antenna and matching network between the transmitter and receiver, thus minimizing the system cost by reducing external components. The transceiver is implemented using 1P6M 55-nm CMOS technology. The die area of the transceiver with DC-DC converter is 1.79 mm². The power consumption of the transmitter and receiver are 6 and 5 mW, respectively, when the output power level of the transmitter is 0 dBm. The noise figure of Rx is up to 6.8 dB with respect to channel frequencies. The phase noise of the ADPLL is -84.7 and -118.9 dBc/Hz at 100 kHz and 1 MHz offset from 2.44 GHz, respectively.

Keywords—ADPLL; BLE; FSK transceiver; low power; SPDT

I. INTRODUCTION

Recently, the demands for internet of things (IoT) are increased in many RF applications such as wearable devices, health care, and sensor networks [1]. The IoT application requires long battery life. Bluetooth low energy (BLE) standard is a good candidate for low power IoT applications with high security [2]. Since the RF transceivers for BLE require small size and low cost, high integrations are essential. Therefore, external components should be minimized such as RF impedance matching, baluns, and transmitter (Tx) and receiver (Rx) switch. Especially, it is beneficial to have a BLE SoC which can interface directly with a single-ended 50-Ω antenna instead of an off-the-shelf ceramic component. The presented mass-produced BLE SoC features such an RF interface and still achieves excellent Rx dynamic range, Tx spurious performance and low power operation.

The BLE operates in the 2.4 GHz industrial, scientific, and medical (ISM) band and uses Gaussian frequency-shift keying (GFSK) with a modulation index $h=0.5$ for signaling. Since GFSK signal has only frequency differences with constant envelope, phase lock loop (PLL) based modulators are widely used instead of IQ modulator [3].

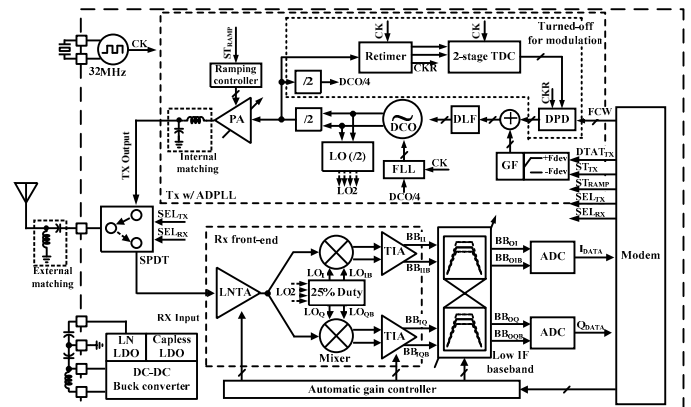


Fig. 1. Block diagram of the proposed BLE transceiver.

The conventional analog PLL is directly modulated by changing division ratio of the feedback divider with sigma-delta modulator (SDM). In that architecture, the power consumption is increased, because additional blocks are needed to implement for the FSK modulation.

In this paper, to implement the low power transceivers, all-digital PLL (ADPLL) frequency synthesizer is adopted for direct GFSK modulator. An ADPLL can easily generate GFSK modulated signals directly by using a digitally controlled oscillator (DCO). Since the direct modulation scheme is possible to generate modulation signals with open loop operation, power consumption of transceiver can be reduced. Many ADPLL blocks for locking are turned off such as retimer, time-to-digital converter (TDC), and digital phase detector (DPD) for transmitting modulated signals as shown in Fig. 1. In addition, the proposed BLE transceiver has integrated single pole double through (SPDT) switch to minimize external components.

II. TRANSCIEVER DESIGN

Fig. 1 shows block diagram of the proposed transceiver for BLE Application. It is composed of a frequency synthesizer using an ADPLL, the dual mode transmitter with class-D RF power amplifier (PA), an integrated SPDT switch, and the power management unit with DC-DC converter in triple-mode and the low drop-outs (LDO). The architectures of Tx and Rx

are an ADPLL based on direct modulation, and low IF down-conversion, respectively. For small area and high efficiency, the inductor-less class D type PA are employed.

The RF front-end of Rx is composed of an inductor-less low-noise trans-conductance amplifier (LNTA), a passive quadrature down-conversion mixer and the trans-impedance amplifiers (TIA). For image rejection, cross coupled complex active-RC band pass filters (BPF) are adopted between variable gain amplifiers (VGA). In order to ease application development, an on-chip DC-DC converter supports battery operation from 1.5 V to 3.6 V. Together with the on-chip SPDT switch that eliminates the need for external RF components between chip and antenna, it enables minimum-sized modules and a fast time-to-market.

III. BUILDING BLOCKS DESIGN

A. SPDT Switch

The SPDT switch provides transmitter and receiver mode selection for time division duplex. As shown in Fig. 2, one antenna can be shared between the transmitter and receiver. An output pad is connected to external matching network which has only two components. To match output impedance of PA additionally and suppress the high order harmonics, simple low-pass filter is comprised of integrated C_{INT} and L_{INT} which can reduce the number of external matching component. The SPDT adopts the body floating technique and pull down switches to improve the insertion loss and isolation as shown in Fig. 2 [4].

B. Transmitter with ADPLL

Fig. 3 shows a block diagram of the proposed ADPLL. It consists of a DCO, dividers, a frequency locked loop (FLL), a retimer, a 2-stage TDC, a DPD, digital loop filter (DLF), Tx modulator with Gaussian filter (GF), and a SDM to enhance a 1-bit resolution of DCO. A retimer of the proposed ADPLL retimes DCO frequency as 32 MHz band. Through a 2-stage TDC, phase difference between retimed DCO frequency and reference clock is converted in digital code and becomes input of DPD. When PLL is on, FLL is firstly operated to put frequency of DCO near to the target frequency. After FLL is locked, through a retimer and 2-stage TDC, a DPD compares frequency control word (FCW) and difference between a series of digital codes, and controls fine capacitor bank of DCO to lock the target frequency [5]. The direct modulation technique is proposed to reduce the current consumption when Tx modulation. After frequency is locked, ST_{TX} becomes high for starting modulation and fractional loop for locking becomes off. By using only GF, DLF, SDM, and DCO, fine cap bank is controlled in accordance with $DATA_{TX}$ and frequency of DCO is modulated. In results, current consumption can be drastically reduced at Tx modulation. Moreover, since the 1-bit frequency resolution of DCO including SDM is around 5 kHz and frequency tolerance of BLE specification when modulation is within ± 75 kHz from carrier frequency, modulation specification can be satisfied even though frequency drifts maximum 5 kHz for the fixed cap bank bit [2]. The PA of transmitter is an inverter-based class-D PA. If the on-resistance of inverter is small enough, then the drain voltage will be a square wave with the amplitude of $V_{DD}/2$ and PA can achieve high efficiency in principle [6].

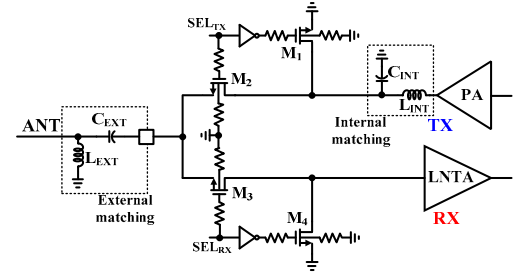


Fig. 2. Schematic of SPDT switch.

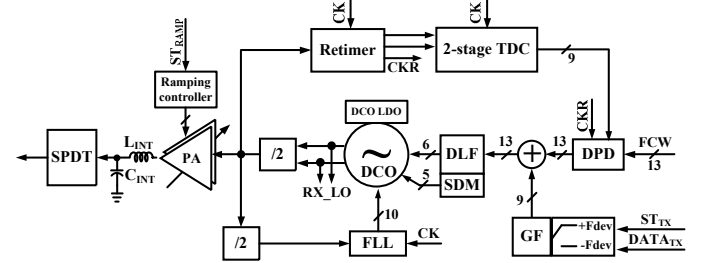


Fig. 3. Block diagram of proposed ADPLL with FSK modulator.

The class-D PA is a single-ended structure and capable of connecting an output of SPDT to antenna directly without on-chip balun. This is important to implement a chip which has extremely small area. A class-D PA is suitable for applications that require nominal output power of 0 dBm. The tuning function of output power is done by dividing the main core of PA into 16 unit cells in parallel. If more cells are active, the overall effective RON is lower, and the output power is increased. This approach enables the output of PA to timely ramp up and down to improve the characteristic of spurious spectral emission. The ramping controller is composed of fully digital logic and is operated when STRAMP is high.

C. Rx RF Front-end

Fig. 4 shows the schematic of RF front-end with 25% duty-cycle current-driven passive mixers. Using 50% duty cycle LO induces I/Q crosstalk. It can be solved by using 25% duty cycle LO. It is composed of a LNTA, passive mixers with the 25% duty generator, a local oscillator (LO), and TIAs. The proposed LNTA in Rx RF front-end is using self-biased inverter type amplifier. The self-biased inverter type amplifier is suitable for low power structure because it has high gain even if the sizes of MOSFETs are small [7]. This type LNTA can operate high speed RF frequency because it doesn't have inner parasitic pole [7]. The input impedance matching of Rx is realized by sharing SPDT with Tx. It guarantees Rx to have a sufficient gain and low noise figure at desired Rx band. In BLE standard, the linearity of mixers is significant issue because of the process of interferers. The passive mixers are used in this work for a low supply voltage operation. A current conveyer structure is used for the LNTA and passive mixer. Therefore, TIA is adopted to convert current to voltage. A 25% duty cycle LO increases the 3-dB conversion gain, which lowers the noise contribution of the mixer comparing to the 50% duty cycle LO [8].

D. Low IF baseband

Fig. 5 shows the block diagram of low IF baseband. The baseband is composed of three stage VGAs, two 2nd order BPFs and two DC offset cancellers (DCOC). Total dynamic

range of baseband is 88 dB. A pseudo differential structure is used for the VGAs for infinite input impedance. The gain steps of VGA1, VGA2, and VGA3 are 4 dB, 2 dB and 1 dB, respectively. The VGAs use differential to single two stage amplifier. The center frequency of the complex BPF is 1 MHz and 3-dB bandwidth is 1.2 MHz. Image frequency rejection ratio is 36 dB. A high pass filter (HPF) is used for the DCOC. An automatic gain control is proposed to control gain of the baseband automatically during preamble duration. However, since the preamble duration is too short, which is 8 μ s in BLE specification, to operate AGC loop, bandwidth of HPF can be controlled by DCOC controller so that bandwidth becomes wide enough during preamble duration to operate AGC loop.

E. DC-DC Converter and LDOs

Fig. 6 shows a block diagram of the proposed triple-mode DC-DC buck converter. The DC-DC converter consists of a start-up oscillator and the triple-mode (PWM, PFM, and retention) controller. Each mode has different characteristics to achieve the wide load range [9]. The PWM mode controller is designed to operate in the active state to provide the good regulation characteristics of V_{OUT} with low output ripple. The retention and PFM mode controller are designed to reduce the switching losses and internal current consumptions as the operation is sleep or stand-by state. The PWM mode is operated under the load of over 2 mA and PFM mode is operated under the load of between 0.5 mA and 2 mA. The retention mode is started when the load current is below 0.5 mA to improve the efficiency. The DC-DC converter operates as a buck converter generating 1.2 V from a 1.5 V to 3 V supply. The operation is defined by the connection of the external 1 μ H inductor (L_1) and 1 μ F capacitor (C_1). The 1.2 V output of the DC-DC converter supplies a bandgap reference (BGR) and 4 on-chip LDOs composed of a capacitor-less LDO and three low-noise LDO with capacitor, each making a 1 V core supply for the different blocks.

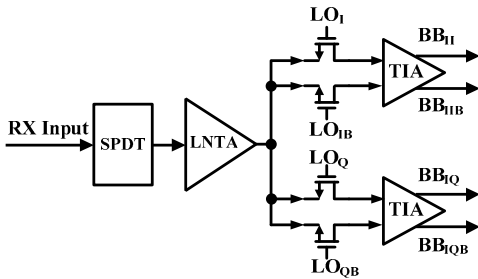


Fig. 4. Schematic of RF front-end with 25% duty-cycle current-driven passive mixers.

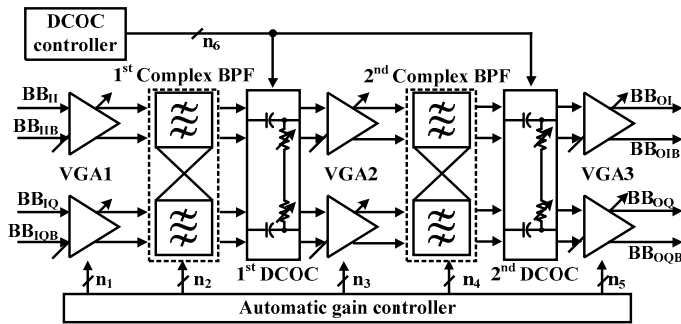


Fig. 5. Block diagram of low IF baseband.

IV. EXPERIMENTAL RESULTS

Fig. 7 shows a micrograph of the proposed transceiver for BLE application which is designed in 1P6M 55-nm CMOS process. The die sizes of transceiver are 1.44×1.14 mm² and that of DC-DC converter is 0.45×0.33 mm² including the ESD protection pad.

Fig. 8 shows the spectrum emission mask of Tx modulation. The measurement result shows the reference emission mask and the modulation result of the proposed BLE transceiver. As can be seen from the results, direct modulation wave of the proposed transceiver meets the specification of the required frequency deviation from 185 kHz to 275 kHz [2]. The spectrum emission mask of Tx is passed when Tx modulation operate [2].

Fig. 9 shows the phase noise measurement results of ADPLL. The phase noise of ADPLL is -84.7 and -118.9 dBc/Hz at 100 kHz and 1 MHz offset from 2.44 GHz, respectively.

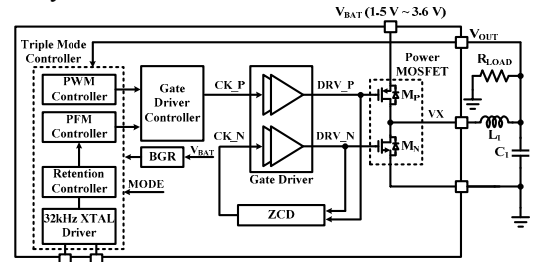


Fig. 6. Block diagram of DC-DC converter in triple-mode.

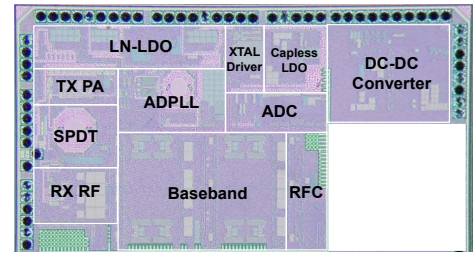


Fig. 7. Microphotograph of the chip.

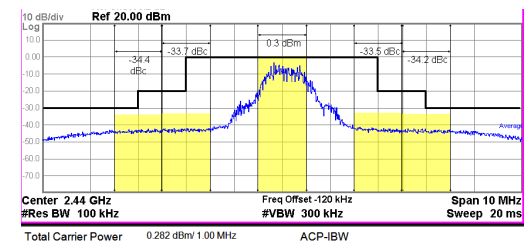


Fig. 8. Spurious emission mask of Tx.

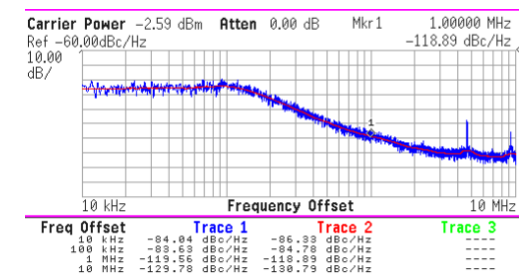


Fig. 9. Phase noise measurement result of ADPLL.

The S11 characteristics of Tx and Rx including SPDT switch as shown Fig. 10 are changed to -14 dB and -18 dB, respectively.

Fig. 11 shows the noise figure and IIP3 measurement results with respect to BLE channel frequencies. The noise figure and IIP is up to 6.8 dB and -18.3 dBm, respectively.

Fig. 12 shows the efficiency of the DC-DC buck converter in triple-mode. It has the highest efficiency and each mode is selected with respect to load current to maximize the efficiency of the proposed DC-DC buck converter. The DC-DC converter operates load current from 1 μ A to 20 mA and has a maximum efficiency of over 89%; this has the best regulation characteristic. Therefore, it is operated as active mode which means the normal operation. The mode is changed from PFM mode to PWM mode at 2.5 mA, because the efficiency of PWM mode is higher than that of PFM mode at 2.5 mA.

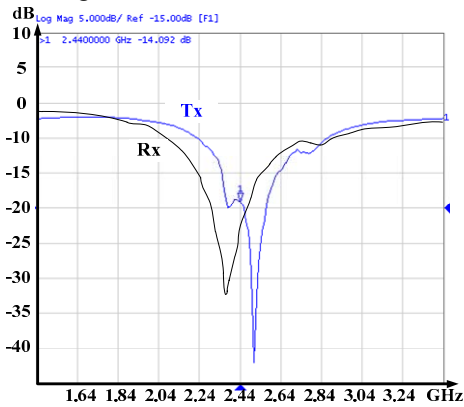


Fig. 10. S11 characteristics of Tx and Rx.

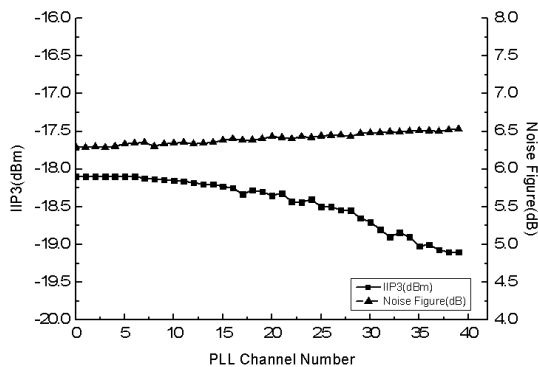


Fig. 11. Noise figure and IIP3 measurement results with respect to BLE channels

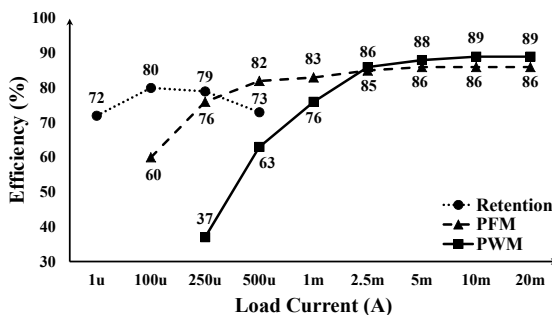


Fig. 12. Efficiency measurement results of the DC-DC buck converter in triple-mode.

Table I shows comparative results of BLE transceivers. Even though this work supports dual mode in Tx mode, the die area is smaller than [10]. However, the die area is larger than [11], this work includes the integrated SPDT switch. The power consumption values of [10] and [11] have been reduced to provide a fair comparison of the power consumption of the Rx and Tx mode. This work has DC-DC converter

V. CONCLUSION

This paper presents a low power FSK transceiver with ADPLL based on direct modulation and integrated SPDT switch for BLE application. To ensure that the proposed low power transceiver can operate at 1 Mbps data rate, GFSK modulation is implemented using an ADPLL with direct modulation technique. The SPDT switch is integrated to share the antenna and matching network between the Tx and Rx, thus minimizing the system cost by reducing external components. The transceiver is implemented using 1P6M 55-nm CMOS technology. The die area of the transceiver with DC-DC converter is 1.79 mm². The power consumption of the Tx and Rx are 6 and 5 mW, respectively, when the output power level of the transmitter is 0 dBm. The noise figure of Rx is up to 6.8 dB with respect to BLE channels. The phase noise of the ADPLL is -84.7 and -118.9 dBc/Hz at 100 kHz and 1 MHz offset from 2.44 GHz, respectively.

TABLE I. COMPARISON OF BLE TRANSCEIVER

	[10]	[11]	This Work
CMOS technology	55 nm	65 nm	55 nm
Supply voltage (V)	3.0	3.0	3.0
TX output power class	0 dBm	0 dBm	0 dBm
TX power consumption (mW)	10.1	10.5	6
	@ 0 dBm	@ 0 dBm	@ 0 dBm
RX architecture	Low-IF	Zero-IF	Low-IF
RX power consumption (mW)	11.2	9.3	5
Noise figure (dB)	5.5	6	*6.8
RX sensitivity (dBm)	-94.5	-94	-94
Area (mm ²)	**2.9	1.1	**1.79

* Total noise figure of receiver

** Include integrated TDD switch

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